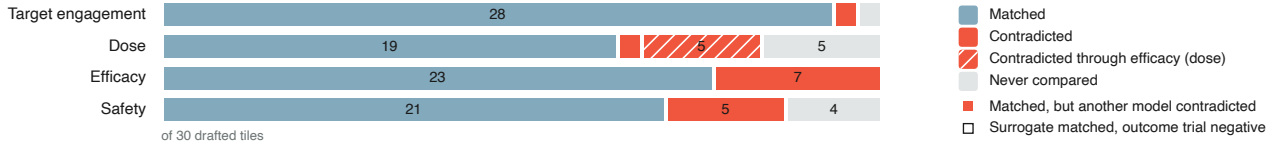
a



Panel b: the top 15 diseases by global DALYs in 2021 (WHO Global Health Estimates), each with the mechanism that had the most programs; drafted tiles only.

b

		Target engagement	Dose	Efficacy	Safety
#1	**type 2 diabetes mellitus** GLP1R agonists (12)	**2011** Cynomolgus monkeys and rat islets (dulaglutide)	**2019** Beagle dog oral bioavailability of semaglutide/SNAC tablets	**2023** GLP1R Gly168Ser genotype, GLP-1 infusion clamp in healthy volunteers	**2016** GLP1R Ala316Thr carriers (human genetics)
#3	**HIV** HIV integrase (INSTI) inhibitors (3)	**2011** HIV-1 replication in cell culture (MT-4/PBMC)	**2013** Protein-adjusted cell-culture potency as plasma trough target	**2012** In-vitro susceptibility of INSTI-resistant HIV-1 (site-directed mutant clones, fold-change)	**2026** Rat and rabbit embryo-fetal development studies Mouse NTD; human signal gone
#4	**NSCLC** EGFR inhibitors (18)	**2017** T790M/sensitising-mutant NSCLC cells and mice	**2019** PC9 brain-metastasis mice; monkey osimertinib PET	**2023** Patient-derived lung cancer organoids EXCLAIM-2 fails primary PFS	**2015** hERG, rat and dog tox, dog telemetry QT prolongation the package cleared
#5	**migraine disorder** CGRP receptor antagonists (4)	**2014** Rhesus capsaicin-induced dermal vasodilation (CIDV)	**2014** Rhesus CIDV PK/PD target Predicted 20 mg; needed 200 mg	**2013** Primate CGRP-induced facial blood flow (rimegepant / BMS-927711)	**2026** Mouse MCAO stroke, gepant pretreatment Animal tox missed MK-3207 liver
#6	**Alzheimer disease** BACE1 inhibitors (4)	**2015** PDAPP mice and cannulated beagle dogs (brain/CSF Aβ)	**2019** Dog CSF Aβ PK/PD scaled to humans Target reached, no benefit	**2019** A673T variant carriers Genetics said protect; drug worsened	**2019** Mice chronically dosed with BACE1 inhibitors Liver injury missed by tox
#7	**asthma** PTGDR2 antagonists (9)	**2016** Human eosinophils and Th2 cells in vitro	**2020** Eosinophil DP2 potency Target reached; exacerbations unchanged	**2014** Allergen challenge Challenge said yes; FEV1 trials missed	**2012** Off-target selectivity panels; animal toxicology unpublished
#8	**breast cancer** CDK4/6 inhibitors (6)	**2016** Human tumour xenografts in mice (phospho-Rb Ser780)	**2017** Abemaciclib xenograft PK/PD (pRb to growth) Brain exposure OK, no response	**2014** Human breast cancer cell-line panel (47 lines) Brain-metastasis trial missed	**2017** Dog cardiovascular telemetry (QTc) with hERG assays Missed liver injury and clots
#13	**schizophrenia** PDE10A inhibitors (5)	**2019** Rat and monkey ketamine-challenge phMRI/EEG reversal	**2019** Rodent occupancy dosing Predicted-adequate exposure, no effect	**2017** Ketamine challenge Challenge passed; phase 2 PANSS missed	**2017** Rat catalepsy, prolactin and glucose vs D2 antagonists EPS in patients at clinical dose
#15	**hepatitis C** HCV NS3/4A protease inhibitors (13)	**2013** Chronically HCV-infected chimpanzee (genotype 1a/1b)	**2009** Replicon EC99 vs rat plasma and liver exposure	**2016** Human hepatocyte chimeric mouse (uPA/SCID) infected with patient HCV genotype 1b	**2016** Rodent and dog repeat-dose toxicology